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TRENDS IN CHEMICAL, ENVIRONMENTAL AND ENERGY TECHNOLOGIES FOR SUSTAINABILITY

Adsorption of methylene blue dye in the thiol-norbornene crosslinked polyvinyl alcohol hydrogel surface: a molecular dynamics study

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Abstract

Methylene blue, a widely used dye in various industries, poses significant environmental and health risks when released into water bodies. The experimental methods, while effective for studying the adsorption of dye on a macroscopic level, lack the atomic-level insights necessary to fully understand the process which can be studied using molecular dynamics simulations. This study investigates the adsorption mechanism of methylene blue on thiol-norbornene crosslinked polyvinyl alcohol (PVA) hydrogel surfaces using MD simulations for the first time. This adsorption behavior is characterized by analyzing hydrogen bonding, radial distribution function (RDF), mean square displacements, and dye removal efficiency. The hydrogen bonding analysis shows consistent stability of the hydrogel network across different pH conditions. The RDF analysis indicates dissociation of the methylene blue molecule, with the cationic part adsorbing onto PVA's hydroxyl groups and the chloride ion exhibits preferential association with the ions in the pH medium. The mean square displacement result suggests a lower diffusion of methylene blue in pH 12.0 which can enhance the rate of adsorption. These findings provide crucial insights into the molecular-level interactions governing dye adsorption on crosslinked PVA hydrogels.

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Highlights

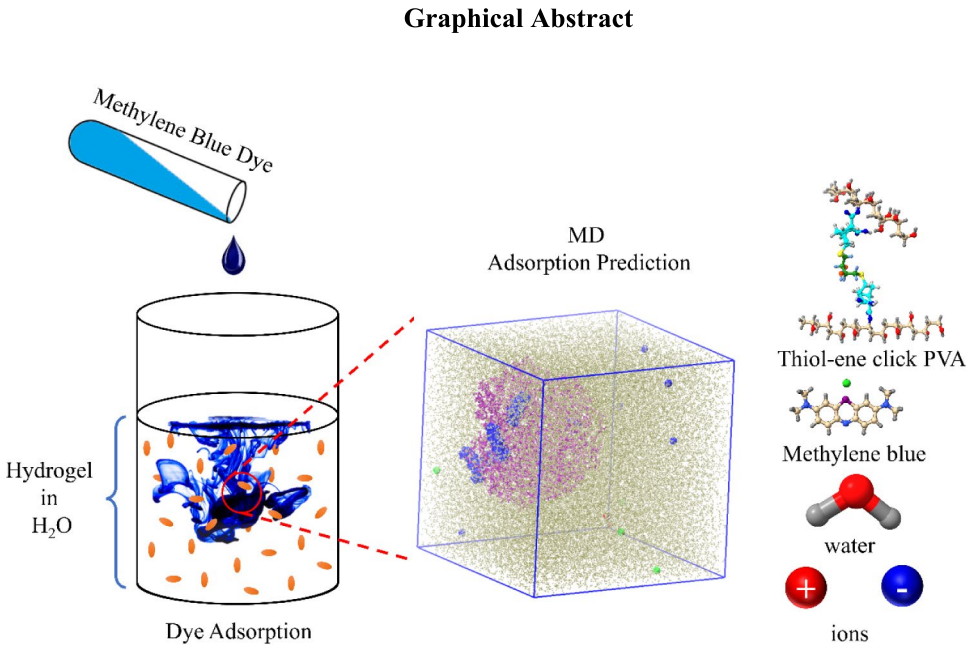
- Molecular dynamics simulation is used to understand the adsorption of methylene blue dye at different pH mediums.
- The thiol-ene "click" polyvinyl alcohol hydrogel is used in the adsorption of methylene blue dye.
- The stability of hydrogel is defined by the hydrogen bonding.
- 40% dye removal is observed with the model in pH 12 at a distance of 0.17 nm.

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Graphical abstract



Keywords Molecular dynamics simulations · Adsorption · Methylene blue · Thiol-ene "click" chemistry · Polyvinyl alcohol · Hydrogel

Introduction

Methylene blue (MB), a cationic thiazine dye, finds extensive application in textile, paper, cosmetic, and medical diagnostic industries (Azhar et al. 2022; Benkhaya et al. 2020; Fito et al. 2023; Novais et al. 2018). Its widespread use and subsequent discharge into aquatic ecosystems pose significant environmental challenges. While utilized as a bactericide and in fish disease treatment, MB transforms into a potent pollutant when present in wastewater (Oladoye et al. 2022). Even at low concentration, its intense coloration adversely impacts water aesthetics, attenuates light penetration crucial for photosynthetic organisms, and disrupts aquatic ecosystems (Hölker et al. 2023). Moreover, MB and its degradation products exhibit toxicity, mutagenicity, and carcinogenicity, presenting substantial risks to aquatic biota and human health through bioaccumulation in the food chain (Oladoye et al. 2022). Exposure can induce various adverse health effects, ranging from dermal irritation and gastrointestinal disturbances to severe conditions like methemoglobinemia, which impairs oxygen transport in blood (Oladoye et al. 2022). The environmental persistence of MB, attributed to its stable aromatic structure, exacerbates these issues, necessitating efficacious removal methodologies. Current remediation strategies encompass adsorption, photocatalytic degradation, biological treatment, and advanced oxidation processes (Aravind et al. 2021; Fong et al. 2020; Hashem et al. 2020; Tichapondwa et al. 2020). Adsorption is particularly favored due to its operational simplicity, high efficiency, and cost-effectiveness. Various adsorbents have been investigated for MB removal, including conventional materials like activated carbon (Ahmad et al. 2020; Misran et al. 2022), clays (El-Naggar et al. 2021; Munir et al. 2020), and zeolites (Abd El Khalk et al. 2021; Mahamud et al. 2023), as well as novel adsorbents derived from agricultural and industrial waste (El-Bery et al. 2022, Hamad and Idrus 2022). However, these methods often encounter limitations such as low selectivity, challenges in material regeneration, and secondary waste generation. In recent years, hydrogel-based adsorption has emerged as a promising alternative, offering advantages such as high water adsorption capacity, tunable porosity, and potential for dye recovery and adsorbent reuse (Hu et al. 2018; Radoor et al. 2020; Teixeira et al. 2017).

The three-dimensional network structure of hydrogel and its tunable properties allow for efficient dye uptake, while their biodegradability ensures minimal environmental impact, making them an attractive and sustainable choice for MB removal from industrial effluents. Hydrogels made from natural or biodegradable polymers like polyvinyl alcohol (PVA) have great potential for removing MB dye from industrial wastewater (Radoor et al. 2020;

Teixeira et al. 2017). These hydrogels can be functionalized with specific functional groups to enhance their affinity toward the cationic dye molecules, enabling efficient dye uptake (Sandeman et al. 2011; Zhao et al. 2024). However, synthesizing PVA hydrogels with optimal properties for MB adsorption remains challenging (Sandeman et al. 2011; Zhao et al. 2024). While physical and enzymatic crosslinking methods offer simplicity and mild reaction conditions, they often result in hydrogels with limited mechanical strength and stability, which can compromise their performance in harsh wastewater environments (Kenawy et al. 2014). Chemical crosslinking strategies, particularly the thiol-ene "click" reaction, have emerged as promising route to synthesize robust and durable hydrogels for adsorption (Yap et al. 2020). The thiol-ene "click" reaction involves the addition of thiol groups to alkenes under mild conditions. This reaction offers several advantages aligned with green chemistry principles, including high efficiency, rapid reaction kinetics in aqueous environments, high selectivity and yields, and the ability to initiate the reaction with visible light, eliminating the need for harmful UV radiation or toxic photo-initiators (Adarsh et al. 2022; Baudis et al. 2016). Furthermore, the resulting thioether linkages are stable and resistant to hydrolysis, ensuring the longevity of the hydrogel in aqueous environments. The thiol-norborene click PVA hydrogel is synthesized and characterized for biomedical applications such as drug delivery and tissue engineering. The thiol-norbornene "click" PVA hydrogel shows low cytotoxicity, stable storage with pyrogallol, and optimized crosslinking with a 60% thiol-to-ene ratio. Norbornene-modified PVA exhibits higher reactivity, while hydrogels show increased swelling due to carboxylate content (Adarsh et al. 2022; Baudis et al. 2016). However, this material is not explored for the adsorption of pollutants. Despite the advantages of thiol-ene "click" crosslinked PVA hydrogels, a critical knowledge gap exists in understanding the molecular-level interactions between the hydrogel network and MB dye molecules. Experimental studies have provided valuable insights into the macroscopic properties and adsorption performance of these hydrogels, but they lack the ability to probe the underlying mechanisms governing dye–hydrogel interactions at the atomic scale to understand the physical adsorption. This knowledge gap hinders the rational design and optimization of hydrogel systems for enhanced pollutant adsorption. To bridge this knowledge gap, the molecular dynamics (MD) simulation provides detailed insights into the conformational changes, hydrogen bonding patterns, and electrostatic interactions between the hydrogel network and pollutant molecules. By simulating the adsorption process at the atomic scale, researchers can elucidate the role of the thiol-ene "click" functional groups and hydrogel architecture in facilitating dye uptake.

Molecular dynamics (MD) simulations are extensively employed to investigate the adsorption behavior of MB on various surfaces. For MB adsorption, simulations provide valuable insights into its interactions with materials such as coal (Huang et al. 2019), kaolinite (Greathouse et al. 2015), graphene (Castillo & Luis 2023), clay (Sellak et al. 2024), and hydrogel (Adeleke & Lasich 2024; Boukoussa et al. 2021; Chen et al. 2022; Oyarce et al. 2022). These studies reveal that MB tends to form stacked conformations on surfaces, primarily driven by van der Waals forces and π–π interactions. The solvent plays a crucial role in the adsorption process, often destabilizing vacuum-stable conformations and promoting hydrophobic interactions. The PVA hydrogel is used to adsorb the bromothymol blue, tetracycline, and metronidazole using MD simulations and Monte Carlo simulations. The simulation result suggests that the tetracycline has the highest adsorption in the PVA hydrogel surface compared to the other materials (Adeleke and Lasich 2024). The zirconium alginate hydrogel fiber is used to adsorb the MB. The MD result shows that 79% of MB is adsorbed in zirconium alginate hydrogel surface (Chen et al. 2022). The silica gel is used to adsorb the MB in MD simulations. The result shows that the 9-nm pore hydrogel is showing maximum adsorption of MB (Boukoussa et al. 2021). Regarding thiol-ene "click" PVA hydrogels, MD simulations have been used to investigate their structural and transport properties. The thiol-ene "click" PVA hydrogel simulations have revealed the crucial role of norbornene in stabilizing PVA polymer chains and contributing to the overall robustness of the hydrogel. The number of hydrogen bonds and swelling ratio between the molecules of polymer and the solvents are both improved by the thiol-ene "click" PVA hydrogel (Das and Kundu 2023). Literature findings indicate that multiple hydrogel models have been applied for MB adsorption. However, the physically crosslinked PVA hydrogel is used to adsorb MB and other pollutants. However, to the best of our knowledge, the chemically crosslinked thiol-ene PVA hydrogel has not been studied focusing on the adsorption behavior of MB.

To overcome this knowledge gap, the MD simulation is used to study the adsorption of MB dye in the thiol-ene "click" PVA hydrogel, which is the objective of this work. The structural information of the stable thiol-ene crosslinked PVA hydrogel has been considered from the literature (Das & Kundu 2023). The MB dye is added to the hydrogel system to understand the adsorption process at the all-atom level. Additionally, the effects of pH variations on the adsorption behavior are explored, providing insights into the underlying mechanisms governing this crucial interaction. After completion of the MD simulations, the structural, transport, and adsorption properties are computed. After completing the MD simulations, a thorough analysis of the structural, transport, and adsorption properties has been

conducted, revealing the fundamental principles that determine the performance of these hydrogels in dye removal applications.

Computational methodology

The molecular structures of vinyl alcohol and MB are constructed using Avogadro open-source package (Hanwell et al. 2012). A polyvinyl alcohol (PVA) segment is generated by replicating the vinyl alcohol unit ten times, which is described in the literature (Das & Kundu 2023). To enhance functionality, norbornene groups are attached to specific hydroxyl positions along the PVA backbone, mitigating potential steric effects. Subsequently, thiol moieties are introduced to the norbornene's unsaturated sites, yielding a thiol-modified PVA variant. Both the norbornene-functionalized and dithiothreitol-incorporated PVA chains are expanded to create extended polymers, each comprising 80 vinyl alcohol monomeric units, which is described in the literature (Das & Kundu 2024, 2025; Deshmukh et al. 2024; Thombre et al. 2024). The experimental evidence suggests that the dithiothreitol acts as a crosslinker (Baudis et al. 2016). To create a thiol-ene crosslinked PVA gel, a couple of norbornene functionalized chains and dithiothreitol functionalized chains are crosslinked to generate the enclosed polymer framework, which is given in Fig. 1.

The three distinct pH mediums, namely, 2.0, 7.0, and 12.0, are created to understand the adsorption of MB. The pH 7.0 is created with the addition of TIP4P (Jorgensen et al. 1983) water to the simulation box. To create pH 2.0 and pH 12.0 environments, appropriate quantities of hydrochloric acid and sodium hydroxide, respectively, are introduced into separate pH 7.0 simulation boxes. The detailed system information is given in Table 1. Furthermore, the MB dye and PVA-NTC polymer framework are subsequently immersed into these pH-adjusted media using the Packmol open-source package (Martínez et al. 2009). The Optimized Potential for Liquid Simulations (OPLS) (Jorgensen et al. 1996) force field is employed to model both bonded and non-bonded interactions of the polymer framework, MB dye, and water molecules. The functional form of the OPLS force field is described in the literature (Jorgensen et al. 1996, Sambasivarao and Acevedo 2009). The force field parameters for hydrochloric acid and sodium hydroxide are taken from existing literature (Bonthuis et al. 2016). The partial charge of each atom is calculated using the Gasteiger method. The dynamics of the system are obtained using the GROMACS-2022.3 open-source package (Bauer et al. 2022).

A computational framework is developed to perform energy minimization, equilibration, and production on these systems used to explore the dynamics of the system. The system is energy minimized by using steepest descent algorithm with 10 kJ/mol/nm as energy threshold cut-off and running for maximum steps until 500,000 in step size of 0.001. The van der Waals cut-off distances are maintained at 1.0 nm, and short-range electrostatic interactions cut-off are maintained at 1.2 nm. The long-range electrostatic interactions are computed using a particle mesh algorithm (Essmann et al. 1995). The system's periodicity is maintained in all three dimensions. The equilibrations are performed using NVT and NPT ensemble. The Leapfrog integrator is used to integrate the equation of motion with a time step of 2 fs. The NVT ensemble of 5 ns, and NPT ensembles for 15 ns. During the equilibration process, the temperature coupling is performed via a velocity rescale (Bussi et al. 2007) approach at 300 K and pressure couples with the Parrinello–Rahman (Parrinello & Rahman 1981) method at 1.013 bar. Linear Constraint Solver (LINCS) (Hess et al. 1997) algorithm permitted to freeze hydrogen-containing moiety vibrations during equilibration phase. The production run is 30 ns under NPT ensemble. The system's temperature is maintained at 300 K using a velocity-rescale thermostat (Bussi et al. 2007). The system's pressure is maintained at 1.013 bar using Parrinello–Rahman barostat (Parrinello & Rahman 1981). The simulation trajectory recorded every 10 ps for statistical average. These trajectories are visualized by using VMD (Humphrey et al. 1996) and Chimera X (Goddard et al. 2018; Meng et al. 2023) packages. The analysis of trajectory is performed with in-house Python scripts.

Analysis of trajectory

Radial distribution function

The radial distribution function (RDF) is a crucial concept in molecular dynamics (MD) simulations that provides insights into the structural organization and intermolecular interactions within a system. It describes the probability of finding particles at a given distance from a reference particle. The mathematical formulae of RDF are given in the literature (Deshmukh et al. 2024). The RDF provides valuable information about the short-range and long-range order, coordination shells, and the presence of specific patterns or structures within the simulated system. It is widely used in the analysis of liquids, solids, and biomolecular systems to understand their structural properties and intermolecular interactions.

The average number of neighboring particles that surround a central particle or atom within a specified cut-off distance is called a coordination number. It provides insights into the local environment and packing arrangements in the simulated system. The mathematical formulae of *CN* are given in the literature (Deshmukh et al. 2024). The coordination number is a useful metric for characterizing the

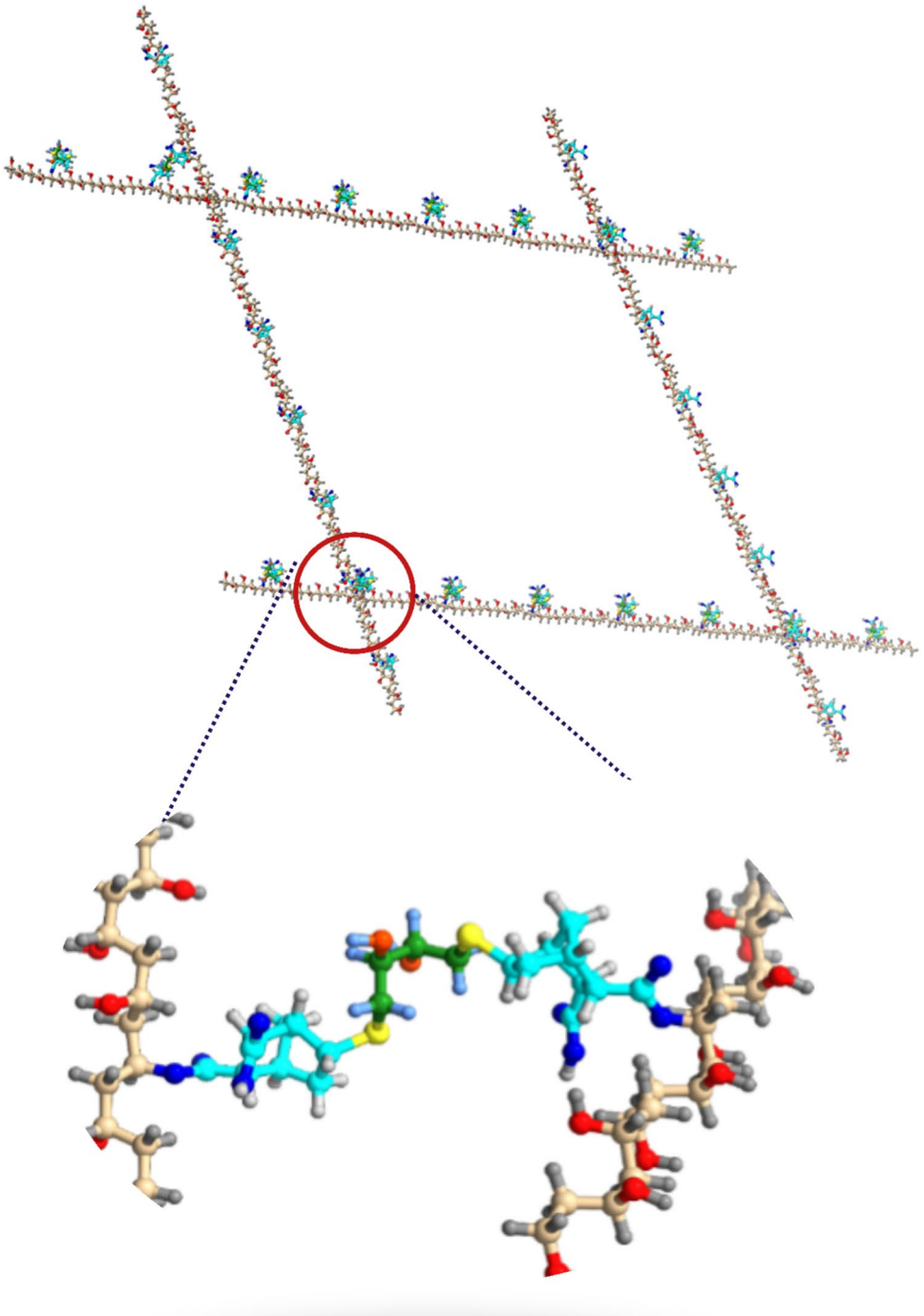


Fig. 1 Initial structure of thiol norbornene crosslinked PVA hydrogel

local structure, coordination shells, and packing efficiency in various systems, such as liquids, solids, and biomolecular assemblies.

Hydrogen bonding (H-bond)

A hydrogen bond is formed when a hydrogen atom covalently bonded to an electronegative atom (donor) interacts with another electronegative atom (acceptor) through an electrostatic attraction. In MD simulations, hydrogen bonds are typically identified based on geometric criteria, such as the distance between the donor and acceptor atoms (r_{DA}) and the angle between the donor–hydrogen bond and the hydrogen–acceptor vector (θ_{DHA}). The mathematical formulae and geometric criteria are given in the literature (Bayles et al. 2021; Ren et al. 2008).

Table 1 Initial system details

System name	pH 2.0	pH 7.0	pH 12.0
Initial box dimension (nm^3)	12.0×12.0×12.0	12.0×12.0×12.0	12.0×12.0×12.0
Final box dimension (nm^3)	8.73×8.73×8.73	8.73×8.73×8.73	8.74×8.74×8.74
Name of hydrogel	PVA-NTC	PVA-NTC	PVA-NTC
Number of atoms in the polymer	6352	6352	6352
Number of water atoms	80,000	80,000	80,000
Number of H_3O^+ atoms	16	–	–
Number of Cl^- atoms	4	–	–
Number of Na^+ atoms	–	–	4
Number of OH^- atoms	–	–	8
Number of methylene blue atoms	195	195	195
Total number of atoms	86,567	86,547	86,559
Final density of the system (kg/m^3)	1010.56	1010.51	1009.96

Mean square displacements

Mean square displacement (MSD) is a measure of the average distance traveled by particles or atoms in a molecular dynamics (MD) simulation over time. The MSD can be defined as

$$MSD = \frac{1}{N}\sum_{i=1}^{N}\left(r_i(t) - r_i(0)\right)^2 \quad (1)$$

where N is the number of particles, and $r_i(t)$ and $r_i(0)$ are the positions of particle i at time t and the initial time, respectively. The MSD slope is characterized by β. It can be defined as

$$\beta = \frac{d(\ln(MSD))}{d(\ln(t))} \quad (2)$$

If $\beta = 1$, then the system is called diffusive. If $\beta < 1$, then the system is called sub-diffusive. If $\beta > 1$, then the system is called super-diffusive behavior.

Dye adsorption study

The percentage of dye removal is a measure of the efficiency of an adsorption process in removing a dye from a solution. To calculate the percentage dye removal in MD simulation, the distance between the dye molecules to the surface molecules of PVA-NTC is computed. The cut-off distance is taken from the other analysis such as RDF and H-bonding. If the dye molecule presents in the cut-off distance, then it is computed as adsorption by the polymer framework. The removal efficiency is mathematically defined using the following formula:

$$Removal(\%) = \left(\frac{c_0 - c_e}{c_0}\right) * 100 \quad (3)$$

where c_0 = initial concentration of the dye in solution and c_e = equilibrium concentration of the dye in solution after adsorption.

Result and discussion

Stability of the hydrogel in the buffer medium

Figure 2 provides a comprehensive depiction of the hydrogen bond formation between the polymer and the solvent and between the polymer chains themselves. Meticulous analysis shows that the mean number of H-bonds formed between individual polymer chains is 164 per frame, consistently across different pH levels (pH 2.0, pH 7.0, and pH 12.0). The polymer-to-polymer H-bonds suggest the self-assembly behavior of the polymer chains. Concurrently, in the polymer to solvents, the mean quantity of H-bonds is 192 per frame. The consistent quantity of H-bonds between the polymer segments and the solvent molecules across different pH conditions implies that the PVA-NTC chain maintains its structural integrity, regardless of the pH environment. Furthermore, the disparity of H-bonds formed between the polymer and water molecules is observed significantly higher than the number of H-bonds formed between individual polymer chains. This phenomenon implies that the polymer has a higher cohesive force, attributed to the water's inherent properties with a higher dielectric constant compared to the polymer segments. The higher dielectric constant of water facilitates stronger hydrogen bond formation with the polymer, as the polar water molecules can effectively screen and stabilize the moieties involved in the hydrogen bonding interactions. In addition to the dielectric constant, the concentration of water molecules and the specific structural arrangement of the polymer segments are identified as influential factors in the construction of H-bonds. The high concentration of water molecules in the system ensures an

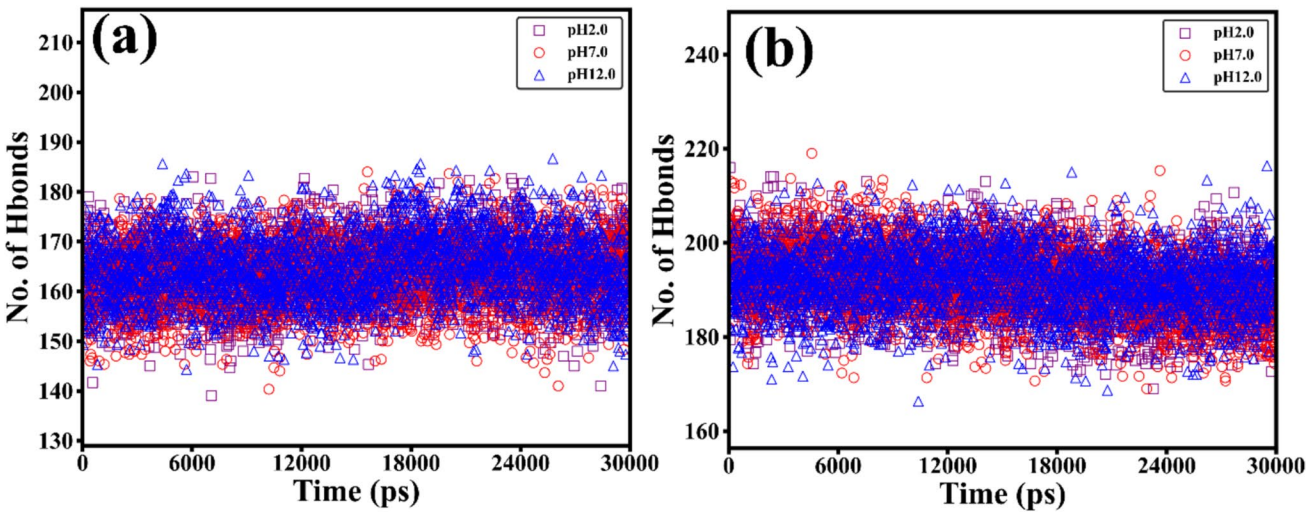


Fig. 2 Average number of H-bonds: **a** polymer to polymer, **b** polymer to solvents

abundance of available hydrogen bond acceptors and donors, thereby enhancing the probability of hydrogen bond formation between the polymer and the solvent. Moreover, the specific conformation and orientation of the polymer segments can either promote or hinder the formation of hydrogen bonds, depending on the accessibility and alignment of the hydrogen bond donor and acceptor groups. The implications of these differences in hydrogen bonding patterns are significant, as they directly impact various properties of the polymer hydrogel. For instance, the increased hydrogen bonding between the polymer and water molecules can enhance the solvent resistance of the hydrogel, as the strong polymer–solvent interactions prevent the penetration and disruption of the hydrogel network by the solvent molecules. Additionally, the hydrogen bonding interactions can influence the thermal properties of the hydrogel, such as its glass transition temperature and thermal stability, as well as its mechanical properties, including tensile strength and Young's modulus (Bayles et al. 2021; Ren et al. 2008). Interestingly, as depicted in Fig. S1, the absence of hydrogen bonds between MB and both the polymer and the solvents is observed. This observation suggests that MB, the dye molecule under investigation, does not form hydrogen bonds with the PVA-NTC hydrogel or the solvent molecules. Instead, it is likely that the dye molecules adsorb onto the polymer surface through alternative mechanisms, such as covalent bonding or electrostatic interactions. The lack of hydrogen bonding between the dye and the hydrogel system may be attributed to the specific chemical structure and electronic properties of MB, which may not facilitate the formation of favorable hydrogen bond interactions with the polymer or the solvent molecules. The elucidation of these hydrogen bonding patterns and their implications is crucial for understanding the fundamental behavior and properties of the thiol-ene crosslinked PVA hydrogel system, as well as its interactions with various adsorbate molecules.

Adsorption of MB in hydrogel

Figure 3a shows the RDF between the sulfur moiety of MB to the chloride ion present in the MB dye. The RDF graph

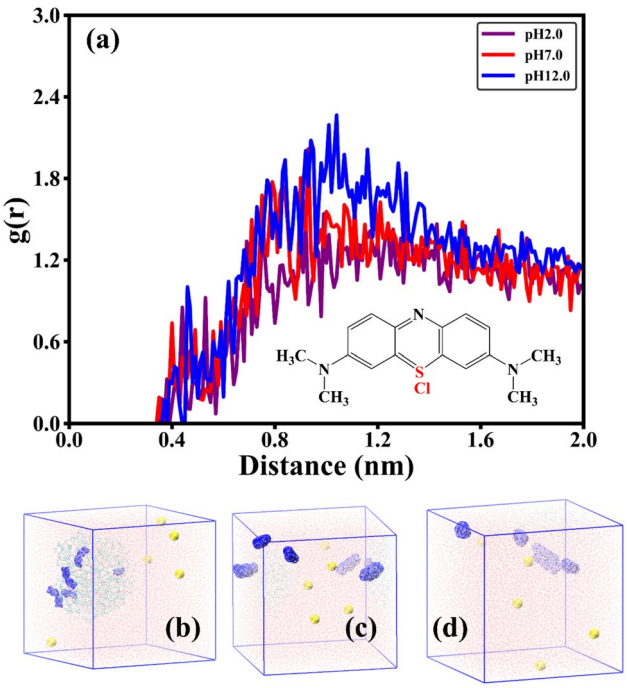


Fig. 3 **a** RDF between the thiol moiety of methylene blue to the chloride ion moiety of methylene blue. **b** Simulation snapshot of MB in pH 2.0. **c** Simulation snapshot of MB in pH 7.0. **d** Simulation snapshot of MB in pH 12.0

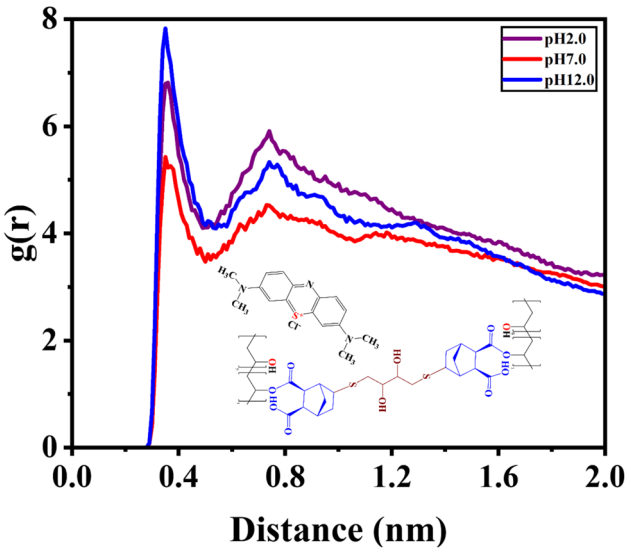


Fig. 4 RDF between the oxygen moiety of the PVA-NTC chain to the thiol moiety of methylene blue

shows the continuous fluctuation in different pH mediums. This signifies that the chloride ion molecules are dissociated from MB molecules and mixed with the solvent which is also suggested in Fig. 3b–d. These simulation snapshots are focused on the MB and chloride ion moiety of MB. These simulation snapshots are clearly visible that the chloride ion of MB is not stuck with other MB moieties which also suggest the RDF discussion. To understand the adsorption process of MB dye, there are several RDFs plotted between the different moieties of PVA-NTC polymer framework to the thiol moiety of MB. Figure 4 shows the RDF between the oxygen moiety of the hydroxyl group of PVA-NTC and the thiol moiety of MB. The first solvation shell is observed at 0.50 nm with a *CN* of 0.52, 0.56, and 0.66 for pH 2.0, pH 7.0 and pH 12.0, respectively. The first solvation shell between the thiol moiety of MB to oxygen moiety of PVA-NTC is observed due to the electrostatic interactions between the moieties. The order of *CN* is observed due to the surrounding conditions. At acidic pH, the surface of the PVA becomes positively charged due to the presence of H^+ ions, which creates an unfavorable repulsion between the cationic MB and the crosslinked polymer. However, as the pH increases, the polymer becomes negatively charged, making it more favorable for attracting the MB molecules. This is reflected in the higher adsorption capacity at higher pH values, which is suggested in the experimental literature. Figure S2 in the Supplementary material shows the RDF between the thiol moiety of MB to the unreacted ene moiety and the thiol moiety of MB to the oxygen moiety present in the norbornene structure. The first solvation shell is observed due to the electrostatic interaction between the moiety. To understand the diffusive behavior of thiol moiety, the MSD has been plotted. Figure 5 illustrates the MSD of

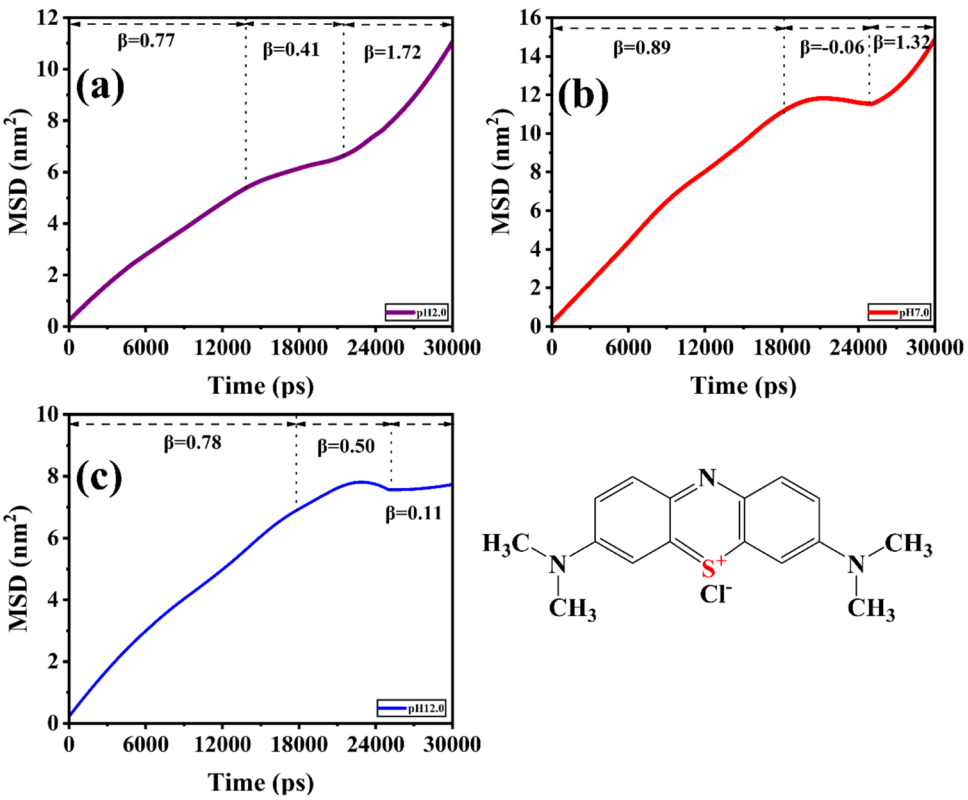


Fig. 5 MSD of thiol moiety of methylene blue: **a** pH 12.0, **b** pH 7.0, **c** pH 12.0

thiol moiety in the MB at different pH mediums. The MSD curve shows two types of diffusive behavior observed: subdiffusive and superdiffusive behavior. Initially, the subdiffusive behavior is observed, which is due to the dye molecules being close to the hydrogel surface where the motion of dye moiety is restricted by the hydrogel molecules. At the end of the dynamics, the superdiffusive behavior is observed which is due to the less statistical averaging. However, the lowest MSD value is observed at pH 12.0, which suggests the restriction of dye molecules by the hydrogel molecules, and further concludes the higher adsorption at higher pH.

To understand the adsorption of chloride ion moiety, several RDFs are plotted between the chloride ion moiety and other components present in the pH medium. Figure 6 shows the RDF between different ions present in the pH medium to the chloride ion of MB dye. Figure 6a shows the RDF between the hydrogen moiety present in the hydroxonium ion and the chloride ion of MB. The first solvation shell is observed at 0.25 nm with a *CN* of 0.03. This RDF peak is observed due to the ionic interactions between the moieties. The second solvation shell is observed at 0.40 nm with a *CN* of 0.14. The second solvation shell is observed due to the ionic interactions between the moieties. The RDFs from Fig. 6a inset, 6b, and 6c inset are shown between the hydrogen atom of water moiety and the chloride ion moiety of MB. The solvation shell is observed at 0.30 nm with a *CN* of 6.62. This solvation shell is observed due to the electrostatic interactions. Due to these minor interactions, the

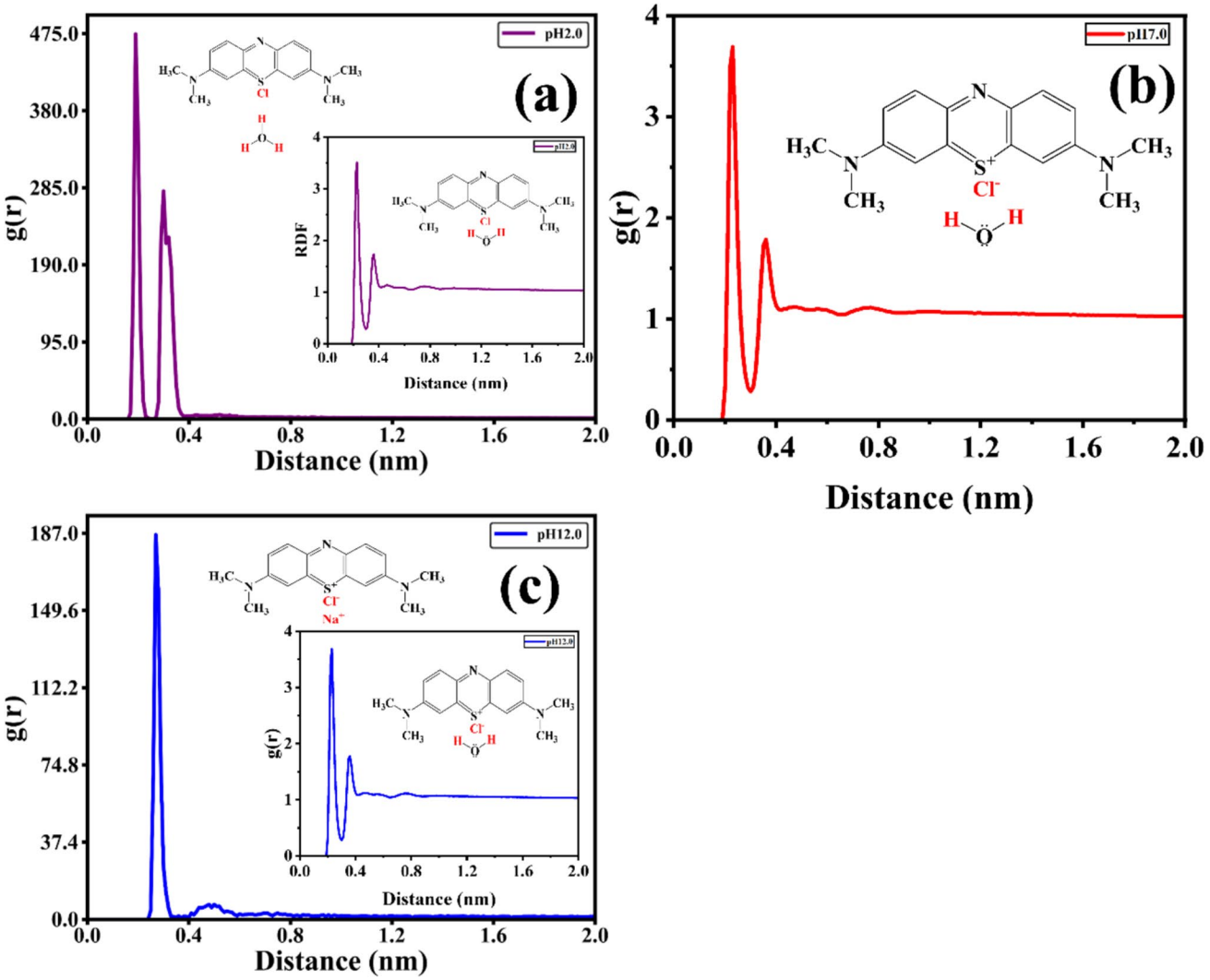


Fig. 6 RDF between the ions present in different mediums to the chloride ion moiety of methylene blue: **a** hydrogen moiety of H_3O^+ ion, (inset) hydrogen moiety of water to the chloride ion moiety of methylene blue; **b** hydrogen moiety of water to the chloride ion moiety of methylene blue; **c** sodium moiety to the chloride ion moiety of methylene blue, (inset) hydrogen moiety of water to the chloride ion moiety of methylene blue

diffusion of chloride ion is affected by the ions present in the pH medium, which is discussed in subsequent sections. Figure 6c shows the RDF between the chloride ion moiety of MB and sodium moiety present in the pH 12.0 medium. The first solvation shell is observed at 0.39 nm with a *CN* of 0.04. This first solvation shell is observed due to the ionic interactions between the moiety. Comparing Fig. 6a and c, the highest RDF value is observed at pH 2.0 medium. This is due to the acidic medium pH; the solutions become positively charged due to the presence of H^+ ions, which creates a favorable attraction force between the chloride ion ions and hydrogen atoms of H_3O^+.

To understand the self-diffusion characteristics of chlorine, the MSD of chloride ion has been plotted. Figure 7 shows the MSD of chloride ion in different pH mediums. Figure 7a shows the MSD of chloride ion in a pH 7.0 medium. It shows a linear behavior throughout the production dynamics. Chloride ion displays super-diffusive characteristics, as evidenced by a logarithmic slope exceeding one. In a pH 7.0 medium, there are no ions present, and chloride ion interacts with only the hydrogen moiety of water with a long-range interaction, as discussed in the previous section. As a result, it moves freely in the pH 7.0 medium showing the linear behavior. Figure 7b shows a MSD of chloride ion in a pH 2.0 medium. Initially, it shows a linear behavior up to 20.9 ns. During this time period, the logarithmic slope value is less than one, which signifies the subdiffusive behavior. During this time period, the movement of chloride ion moiety is restricted by the ions present in the pH 2.0 medium. After that time period, the MSD slope decreases until the end of the dynamics. This decrease in MSD slope is observed due to the chloride ion trying to free from the ions' pocket. Figure 7c shows the MSD of chloride ion moiety in pH 12.0 medium. Initially, it shows a linear behavior until 10.8 ns, where the logarithmic slope value is less than one. It signifies subdiffusive behavior which is due to interactions between the ions present in the pH 12.0 medium. After that, the MSD slope decreases until 24.1 ns. Thereafter, the MSD shows superdiffusive behavior until the end of the dynamics, which is suggested by the logarithmic slope. During this time period, the chloride ion moiety breaks the ion traps. As a result, the motion of chloride ion ions increases and it shows superdiffusive behavior. Comparing the MSD of chloride ion in different mediums, it shows that the highest MSD is observed at pH 7.0. This is observed due to the non-ionic interaction between the chloride ion chloride ion and solution. The ions in pH medium reduce the motion of chloride ion due to their strong interactions. The second highest MSD is observed in pH 2.0 medium where concentration of H_3O^+ ions is higher. These ions attract chloride ion moiety strongly, and their atomic mass

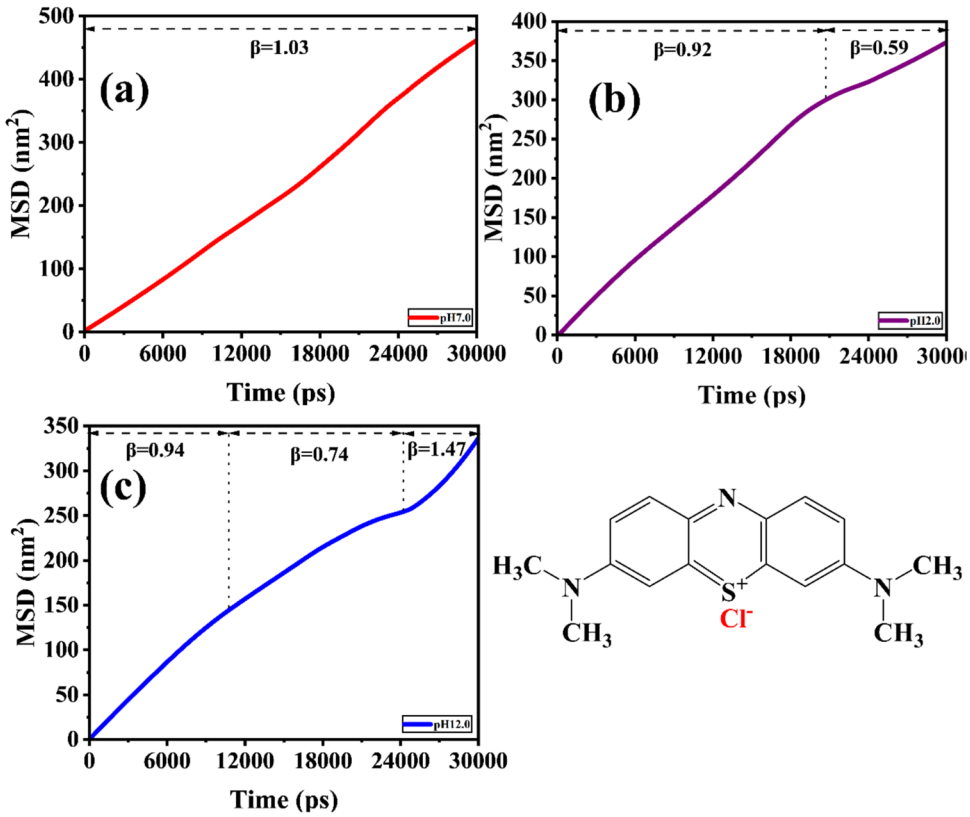


Fig. 7 MSD of chloride ion moiety present in methylene blue: **a** pH 12.0, **b** pH 7.0, **c** pH 12.0

is lighter compared to the sodium ions in the base medium. These ions reduce the MSD value compared to the pH 7.0 medium, and lightweight H_3O^+ ions give higher MSD compared to the pH 12.0 medium.

To understand the further adsorption characteristics, the percentage removal of dye has been studied. The MB is adsorbed by the physisorption (Makrigianni et al. 2015; Radoor et al. 2020) and chemisorption (Rehman et al. 2018). Figure 8 shows the simulation snapshots at the end of the 30-ns production run in different pH mediums. The simulation snapshots clearly show that the MB molecules are adsorbing in the PVA-NTC hydrogel surface. To better understand the adsorption process, the distance between MB molecule and the different moieties of PVA-NTC surface is computed. There are two distances observed in between the PVA-NTC surface and MB molecules. The first distance between the PVA-NTC surface and MB molecules is 0.17 nm. At 0.17 nm, the percentage of dye removal is given in Fig. 9a. The order of removal is pH 2.0 = pH 7.0 < pH 12.0. The lowest dye removal is observed at pH 2.0. At acidic pH, the surface of the PVA becomes positively charged due to the presence of H^+ ions, which creates an unfavorable repulsion between the cationic MB dye and the gel. Due to this reason, pH 2.0 is showing the lowest dye removal percentage. The highest percentage of removal is observed in a pH 12.0 medium. In the base medium, the PVA-NTC membrane becomes negatively charged, which is due to the hydroxyl moiety present in the solution. This makes it more favorable

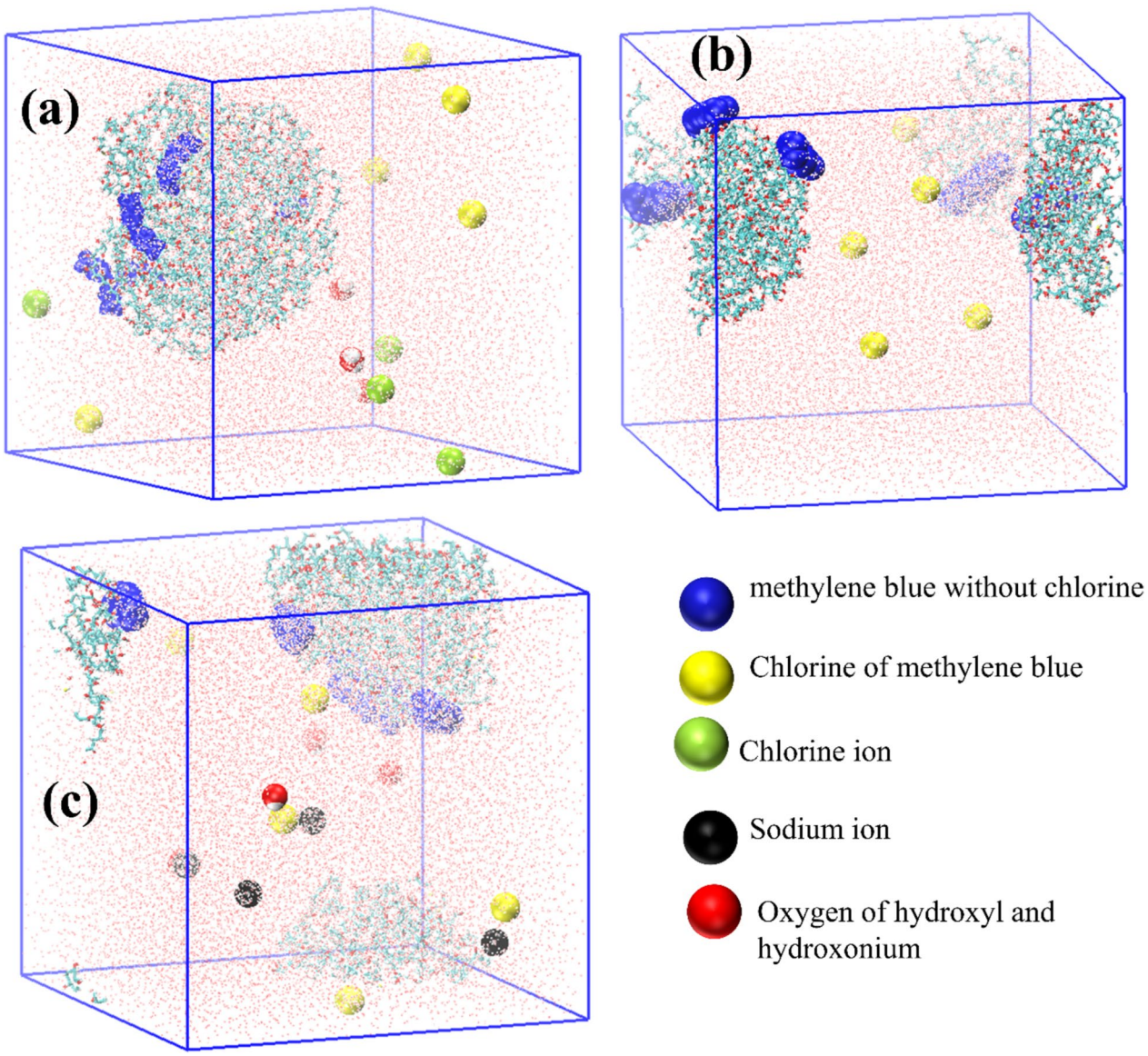


Fig. 8 Final simulation snapshot: **a** pH 2.0, **b** pH 7.0, **c** pH 12.0

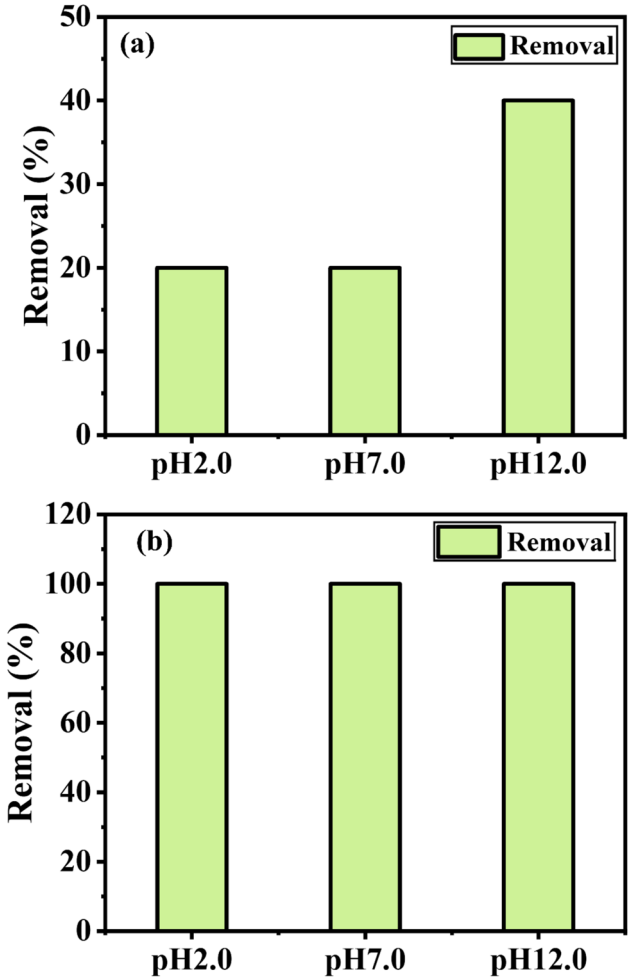


Fig. 9 Percentage removal of methylene blue dye using the PVA-NTC framework: **a** the cut-off distance 0.170 nm, **b** the cut-off distance 0.40 nm

for the chemical interactions to the MB adsorption, which is also validated by the experimental evidence (Meraz Melo et al. 2024; Radoor et al. 2020). Another distance between the MB moiety and PVA-NTC surface moieties is 0.40 nm. At 0.40 nm, the percentage removal of MB using polymer frameworks is shown in Fig. 9b. The dye is 100% removed in each medium. This suggests that the van der Waals and π–π interactions are key to the removal of dye using the PVA-NTC hydrogel framework.

Conclusion

The all-atom MD simulation examines the adsorption of MB dye on the thiol-ene PVA hydrogel surface. Key adsorption characteristics include hydrogen bonding, radial distribution functions (RDF), mean square displacement (MSD), and dye removal. Hydrogen bonding results show that the average number of hydrogen bonds between the polymer chains and between the polymer and solvent remains stable, indicating structural stability of the hydrogel network under various pH conditions. RDF analysis shows that the cationic part of MB dissociates from chloride ions. Further RDF analysis reveals that the cationic part of MB interacts most strongly with the hydroxyl groups of PVA-NTC via electrostatic interactions, while chloride ions tend to bind with ions in the pH medium, highlighting the impact of solution chemistry on adsorption. MSD results indicate that diffusion is lowest in pH 12.0, suggesting a higher rate of dye adsorption in this environment. Both bonded and non-bonded interactions are important for MB adsorption. This study enhances understanding of dye adsorption at the molecular level and offers a basis for optimizing PVA hydrogels for water purification. Future research could explore how varying crosslinking densities, polymer compositions, and environmental factors affect dye adsorption efficiency.

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Declarations

Ethical approval Not applicable.

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